



4.3 Bivariate data and correlation

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A scatter diagram visualises the relationship between two numerical variables. Pearson's product-moment correlation coefficient r ($-1 \leq r \leq 1$) measures the strength and direction of a LINEAR association; values close to ± 1 indicate strong linear correlation, values close to 0 indicate weak/no linear correlation. The least-squares regression line $y = ax + b$, found by technology, can predict y from x within the range of the data (interpolation); predicting outside this range (extrapolation) is unreliable. Correlation does not imply causation.

Worked example: follow the reasoning

Study hours (x) and test scores (y) for 5 students give $r = 0.92$ and regression line $y = 4.5x + 50$. Predict the score for a student who studied 6 hours, given the data ranges from 2 to 9 hours, and comment on reliability.
Solution: $y = 4.5(6) + 50 = 77$. Since $x = 6$ lies within the data range (2 to 9), this is interpolation, and the prediction is likely reliable given the strong positive correlation ($r = 0.92$).

Quick check · multiple choice

A scatter plot shows points scattered with almost no discernible pattern. Which value of r is most likely?

- A 0.95
- B -0.89
- C 0.03
- D 0.5

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A researcher records the age (years, x) and price (\$1000s, y) of 8 used cars of the same model, finding $r = -0.87$ and regression line $y = -1.2x + 22$. (a) Interpret the value of r in context. (b) Use the regression line to estimate the price of a 5-year-old car. (c) Explain why it would be inappropriate to use this line to estimate the price of a 25-year-old car. [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $r = -0.87$ indicates a strong negative linear correlation: as a car's age increases, its price tends to decrease. (b) $y = -1.2(5) + 22 = 16$, i.e. an estimated price of \$16,000. (c) $x = 25$ lies far outside the range of ages in the original data (extrapolation); the linear relationship may not hold for much older cars (e.g. the model would eventually predict a negative, impossible price), making the estimate unreliable.

Quick check answer

C. A value near 0 indicates essentially no linear relationship, consistent with a scattered plot.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.